

Portfolio Value-at-Risk Engine

Model Overview and Framework Documentation

Risk Modelling — Trade Republic

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1 Introduction

This document provides an overview of the `portfolio_vaR` engine implemented in `risk_analytics.risk_engine` (file: `risk_pylibrary/risk_analytics/risk_engines.py`).

The engine computes one-step-ahead (or multi-day) portfolio **Value-at-Risk (VaR)** and **Conditional Value-at-Risk (CVaR, Expected Shortfall)** for a weighted asset portfolio. Five distinct risk methodologies are available, selected via the `engine` parameter.

2 Code Flowchart

Figure 1 shows the full execution path of `portfolio_vaR()`, from input validation through engine dispatch to the final output DataFrame.

3 Notation

Symbol	Description
$r_{i,t}$	Daily return of asset i at time t
w_i	Portfolio weight of asset i , $\sum_i w_i = 1$
r_t^{pf}	Portfolio return: $r_t^{pf} = \sum_i w_i r_{i,t}$
T	Lookback window (default: 250 trading days)
H	Holding period (default: 1 day)
α	Loss probability level, e.g. $\alpha \in \{0.05, 0.01, 0.001\}$
λ	EWMA decay factor (default: 0.94)
ν	Student- t degrees of freedom

4 Available Engines

engine key	Model	Distribution options
hs	Historical Simulation	non-parametric
ewma	EWMA Volatility (parametric)	Normal, Student- t
gjr	GJR-GARCH(1,1,1)	Normal, Student- t
egarch	EGARCH(1,1,1)	Normal, Student- t
mc	Multivariate Monte Carlo	Normal, Student- t

5 Common Output Format

All engines return a `pandas.DataFrame` indexed by the value date with columns named:

$$\begin{array}{ll} \text{var} \underbrace{N}_{1000(1-\alpha)} - \underbrace{H}_d & \text{(VaR at confidence level } 1 - \alpha, H\text{-day horizon)} \\ \text{cvar} \underbrace{N}_{1000(1-\alpha)} - \underbrace{H}_d & \text{(CVaR at confidence level } 1 - \alpha, H\text{-day horizon)} \end{array}$$

Examples with default quantiles $\alpha \in \{0.05, 0.01, 0.001\}$ and $H = 1$:

- `var950_1d` — 95% 1-day VaR
- `var990_1d` — 99% 1-day VaR
- `var999_1d` — 99.9% 1-day VaR
- `cvar950_1d` — 95% 1-day CVaR (Expected Shortfall)

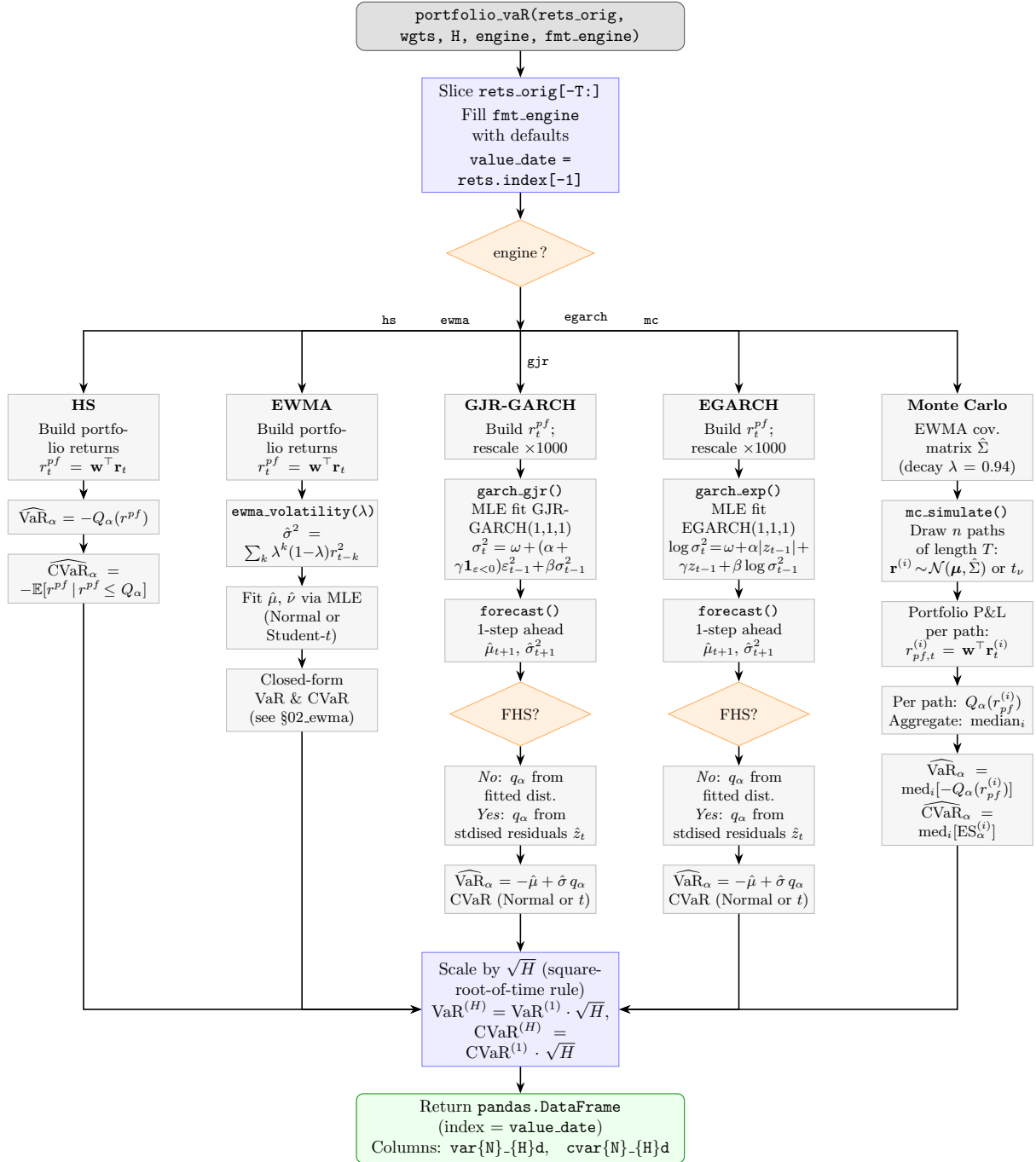


Figure 1: Execution flowchart of `portfolio_var()` in `risk_analytics/risk_engines.py`. The central GJR-GARCH and EGARCH branches share the same VaR/CVaR calculation logic via `calculate_var_garch()`.

6 Holding-Period Scaling

All engines scale 1-day estimates to a H -day horizon using the square-root-of-time rule:

$$\text{VaR}_\alpha^{(H)} = \text{VaR}_\alpha^{(1)} \cdot \sqrt{H} \quad \text{CVaR}_\alpha^{(H)} = \text{CVaR}_\alpha^{(1)} \cdot \sqrt{H} \quad (1)$$

This assumes i.i.d. daily returns and is an approximation for correlated (GARCH-type) processes.

7 Model Cross-Reference

Engine	Parametric tail	Volatility clustering	Multivariate	Non-parametric tail
HS	–	–	–	✓
EWMA	✓	–	–	–
GJR	✓	✓	–	optional (FHS)
EGARCH	✓	✓	–	optional (FHS)
MC	✓	–	✓	–

8 Detailed Model Descriptions

Each engine is documented in a dedicated file within this directory:

- `model_desc/hs/01_historical_simulation.tex`
- `model_desc/ewma/02_ewma.tex`
- `model_desc/gjr_garch/03_gjr_garch.tex`
- `model_desc/egarch/04_egarch.tex`
- `model_desc/mc/05_monte_carlo.tex`